ISEP

Segurança de Sistemas e Informação (SEGSI)
PL01
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Exercise 1: Point-to-Point Protocol (PPP)

PPP is a legacy protocol that can still be found in several implementations. It was used to authenticate before a system. It can use Challenge Authentication or Password Authentication. Take the PPPoE Packet Capture from Moodle and answer the following questions:

1. What kind of authentication is used in the PPP protocol?
2. What are the steps for authenticating the user?
3. Identify the ”secret” password.
4. Imagine you are a telecom provider and you provide this authentication method to your clients, what are your security considerations?

Exercise 2: Challenge Authentication

The Challenge authentication protocol, albeit having several implementations, proposes a set of steps to avoid sharing the password in clear text. However, it is not secure in its default form. Take the Radius Simple Protocol file example from Moodle and answer the following questions:

1. Describe the Radius protocol that you see (What it does, what purpose it serves and its security considerations).
2. Describe the Challenge Authentication Principle.
3. Crack the credential using any password cracker such as JohnTheRipper[1] or Hashcat[2].
4. While it is cracking comment on the resilience of the cipher.

[1] https://github.com/openwall/john
[2] https://github.com/hashcat/hashcat

Exercise 3: LightWeigt Directory Access Protocol (LDAP)

LDAP is a simplistic protocol that is still widely used in corporate Networks. Its simplicity and integration with additional values provide enterprises with scalability and malleability to achieve company goals. It is often forgotten to secure it. Take the LDAP capture file and answer the following questions.

1. What is LDAP Simple Bind? And Anonymous Bind?
2. Name several types of authentications within LDAP.
3. What is LDAP Sign and Seal?
4. What is the advantage of using LDAP with Kerberos authentication?

Exercise 4: NTLM LDAP

LDAP can support several authentication methods. The following is a challenge for you to test your understanding of the protocol. Load NTLM LDAP and answer the following questions:

1. What is the user logging in?
2. Explain the different interactions of the protocol.
3. Extract all pertinent data and crack the password.